

Base Station Antenna Rapid Test Equipment

Antennas that work as soon as they are installed. Damaged antennas back in service, fast.

Base station antennas deliver the last mile of wireless service. Mounted on towers and rooftops, they are the parts of the network most exposed to natural disasters and incidents, and the first priority in any network repair.

ANTestify

Dongguan Voyager Telecom Equipment CO., LTD.

Faster emergency repair and restoration of communication networks

01 Preparedness

Routine operation / before flood season

Keep emergency antenna stock ready for use

- 100% inspection of newly delivered, repaired and remanufactured antennas before they enter the warehouse, so faulty spares are never stocked.
- Sampling inspection of key spares in stock before flood season, to catch hidden damage from long storage.

02 Emergency repair

0–72 hours

Antennas that work as soon as they are installed

- The vehicle-mounted testing platform tests antennas quickly on site.
- Instead of finding a fault after the antenna is on the tower, confirm on the ground that it works: the most direct way to save time and money.

03 Rebuilding

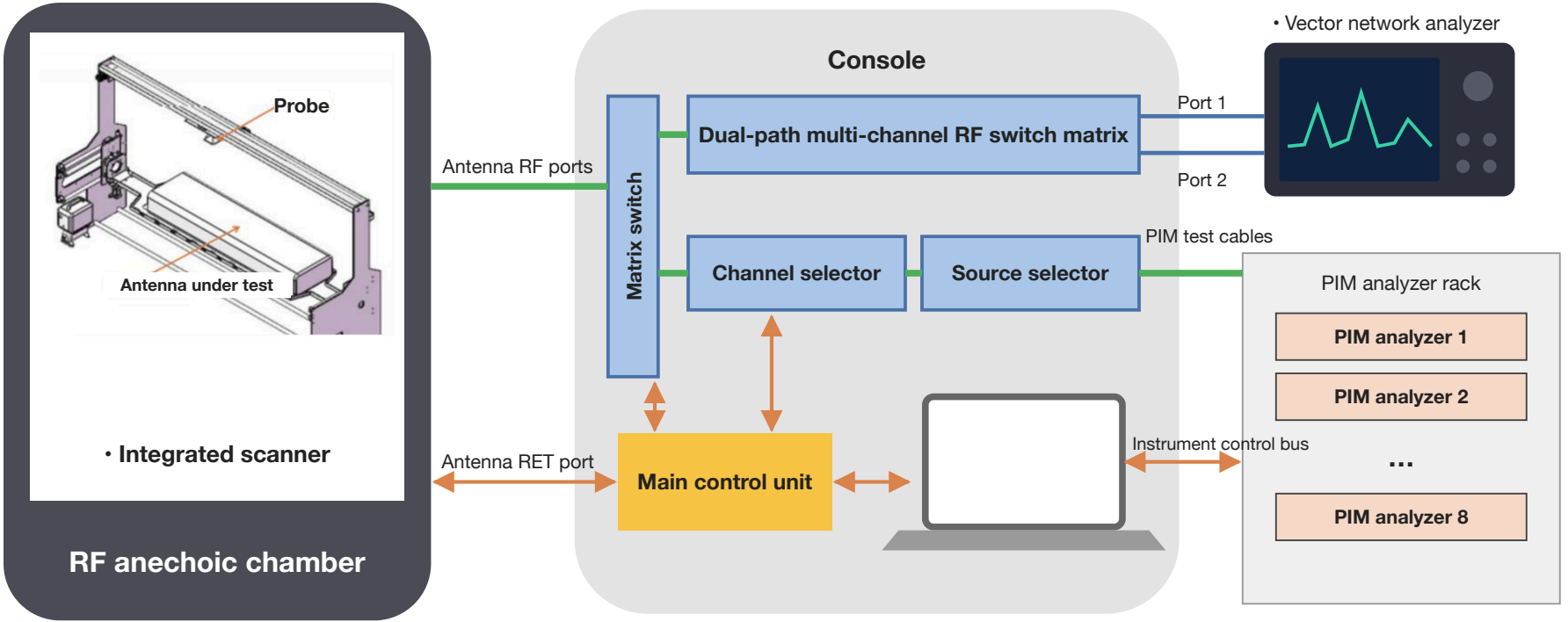
Weeks to months

Fast repair of damaged antennas

Testing on site at material collection points or temporary storage yards, with a three-way verdict:

- **Usable (green)**: all parameters within tolerance, good pattern consistency → back to stock or redeployed nearby
- **Repairable (yellow)**: repairable local defects (connectors, phase shifters, RET modules) → back to stock after repair and retest
- **Scrap (red)**: structural damage or unrecoverable performance → disposed of under asset procedures

Automatic test system = anechoic chamber + instruments + switch matrix + control system



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Circuit, PIM, radiation and RET interface parameters

Category	Parameter	Specification
Circuit	Frequency range	100 MHz – 8 GHz
	VSWR measurement accuracy	±0.01
	Isolation measurement accuracy	±1 dB
	Coupling measurement accuracy	±0.1 dB
PIM	External PIM analyzers	≤ 8
	Signal chain residual PIM	≤ –160 dBc
Radiation	Gain measurement accuracy	0.1 dB
	Horizontal / vertical beamwidth	0.5°
	Electrical downtilt	0.5°
	Cross-polar discrimination	1 dB
	Front-to-back ratio	1 dB
RET interface	Interface type	1. RF: OOK-modulated signalling 2. RS-485: 8-pin connector to IEC 60130-9
	RET protocol	AISG 2.0 / 3GPP
	RAE protocol	AISG ES-RAE v2.2.0

➢ Full-stack in-house R&D

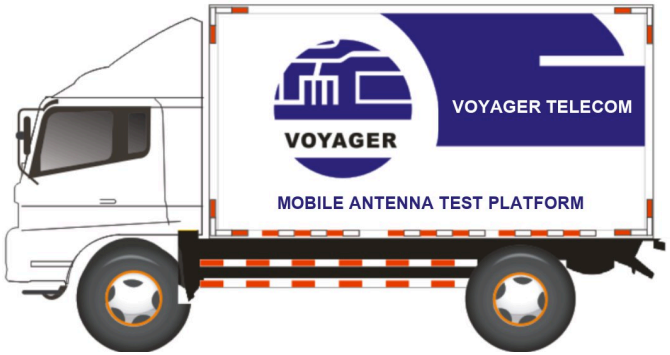
- 3 invention patent applications, including a PCT international application
- 1 utility model patent application
- 9 registered software copyrights

➢ Test process

- Identifies the antenna model and loads the test script automatically.
- Switches the ports under test and sets the electrical downtilt automatically.
- Generates the test report and uploads the test data automatically.

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Vehicle-Mounted Testing Platform: antennas that work as soon as they are installed



➢ Features

- **Complete test coverage.** Tests the antenna's circuit parameters and quickly measures its radiation parameters.
- **Mobile and flexible.** The vehicle-mounted RF anechoic chamber is built with refrigerated-truck body techniques for insulation and sun protection. It works in the field and suits pre-installation testing and rapid network rebuilding after disasters.
- **Five-stage vibration isolation.** Air suspension on the chassis plus vibration-damped mounting of the scanner, console and instruments allow travel on rural roads.
- **On-board power and air conditioning.** The field model works independently for 10 hours or more.

Model	TTC3000-C4G	TTC3000-C8G	TTC3000-U4G	TTC3000-U8G	TTC3000-W4G	TTC3000-W8G
Application	Urban (external power)		Suburban (on-board power)		Field (on-board power + A/C)	
Vehicle	Light truck	Van	Medium truck	Medium bus	Heavy truck	Coach

Integrated Testing System: fast repair of damaged antennas



➢ Features

- **Low cost to build and run.** Typically 10–30% of the price of conventional systems. Works at 25 ± 5 °C, and users maintain and calibrate it themselves, keeping running costs low.
- **Fully automatic testing.** All electrical parameters at one station: a 4+4 antenna in about 20 minutes and a 4+4+8+8 antenna in about 40 minutes (under 2 minutes per port on average), 10–50 times the efficiency of conventional methods.
- **Easy operation.** No alignment to the antenna phase centre, no manual port changes, and the electrical downtilt is set automatically.
- **Automatic report upload.** Test data stay available for the whole product life cycle.

Model	TTC3000-B4G	TTC3000-B8G	TTC3000-H4G	TTC3000-H8G	TTC3000-R4G	TTC3000-R8G
Functions	RET interface + circuit parameters		RET interface + circuit + PIM parameters		RET interface + circuit + PIM + radiation parameters	
Band	600 MHz–4 GHz	600 MHz–8 GHz	600 MHz–4 GHz	600 MHz–8 GHz	600 MHz–4 GHz	600 MHz–8 GHz

Typically 10–30% of the price of conventional systems, all electrical parameters at one station, 10–50 times the efficiency

Criterion	Indoor chamber + multi-probe spherical system	Movable chamber + cylindrical near-field automatic system	Customer value
Construction cost	• Requires a dedicated indoor anechoic chamber • Complex system structure, high cost	• No indoor chamber needed: integrated movable chamber • Simple structure, easy installation and commissioning	**Low construction cost**
Accurate radiation measurement	• Short time per port, long time per antenna • Full radiation measurement of a 24-port antenna: ≥ 3 days	• Slightly longer per port, short time per antenna • Full radiation measurement of a 24-port antenna: < 10 hours	**Fast and accurate**
Rapid full-antenna performance test	• Radiation parameters only	• All electrical parameters at one station • 4+4 antenna in about 20 minutes; 4+4+8+8 antenna in about 40 minutes (under 2 minutes per port on average)	**Much lower test cost**
Automation	• Alignment to the antenna phase centre • Manual port changes • Manual electrical downtilt setting	• No phase-centre alignment • Ports switched automatically • Electrical downtilt set automatically	**Easy operation**
Test report management	• Test data compiled by hand • Pass/fail judged by hand	• Near-field data collected and pattern parameters calculated automatically • Results judged automatically against preset acceptance criteria	**Objective and consistent**
Operation and maintenance	• Strict climate control: 22 ± 2 °C • Annual calibration by the vendor	• Works at room temperature: 25 ± 5 °C • Maintained and calibrated by the user	**Low O&M cost**

Gain measurement capability of the rapid test equipment passed a CTTL measurement audit



CAICT 中国信通院

中国认可
能力验证
PROFICIENCY TESTING
CNAS PT0090

报告编号：CTTL2026MX1007

测量审核报告

移动通信基站天线增益测量能力验证（一对一）

中 国 泰 尔 实 验 室
中国信息通信研究院

2026 年 7 月 8 日

➢ Purpose and method:

Stable low-band and high-band antennas were measured, and the audited laboratory's results were compared with the robust average of data from China's major testing laboratories to verify consistency.

➢ Technical basis:

- GB/T 9410: general specification for mobile communication antennas
- YD/T 2867: multi-band passive base station antennas
- YD/T 2868: measurement methods for passive antennas

➢ Result:

The gain measured by the Vehicle-Mounted Testing Platform was consistent with the results of mainstream laboratories. Detailed data are given in the Result Report issued by China Telecommunication Technology Labs (CTTL, part of CAICT) on 8 July 2026.

China Mobile Hainan: antenna repair and refurbishment with our new equipment

Since 2024, **China Mobile Hainan** has commissioned Voyager to test, repair and refurbish decommissioned and faulty antennas end to end with our new equipment. A standardised repair process fully restores every technical specification to the strict standard required for network re-entry, replacing the habit of buying new with a model of repair instead of replace.

01 Idle assets revitalised

Decommissioned and faulty equipment regains its value through technical means, raising fixed-asset utilisation and reducing idle resources and waste.

02 Procurement costs reduced

Refurbishment costs only 30–50% of buying new, cutting hardware investment in network build-out and operation and improving cash flow.

03 Green development practised

Less electronic waste and pollution and lower manufacturing emissions: a circular-economy model that supports corporate social responsibility.